RetailGPT

Complete RAG Knowledge Base

Superstore Retail Intelligence Document

Metric	Value
Total Transactions	9,800
Total Revenue	$2,261,536.78
Unique Orders	4,922
Unique Customers	793
Unique Products	1,849
Average Order Value	$459.48
Product Categories	3
Sub-Categories	17
US States Covered	49
Cities Served	529

Version 1.0 | RetailGPT Project | Superstore Dataset

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1. Business Overview & Dataset Description

About the Superstore Dataset

The RetailGPT system is built on the classic Superstore retail dataset — a widely used, industry-standard dataset representing a fictional US-based retail company selling products across three major categories: Furniture, Office Supplies, and Technology. The dataset covers multi-year transactional sales data across all four US geographic regions and 49 states, making it ideal for retail analytics, AI-powered Q&A;, and business intelligence workloads.

Dataset Snapshot

Field	Details
Dataset Name	Superstore Sales — Retail Transactions
Total Rows (train.csv)	9,800 transactions
Columns	18 fields per row
Date Range	2015 – 2018 (multi-year)
Geography	United States — 49 states, 531 cities
Categories	Furniture, Office Supplies, Technology
Sub-Categories	17 product sub-categories
Customer Segments	Consumer, Corporate, Home Office
Shipping Modes	Standard Class, Second Class, First Class, Same Day
Total Revenue	$2,261,536.78
Avg. Sale Value	$230.77
Min. Sale Value	$0.44
Max. Sale Value	$22,638.48

Column Definitions

Column	Type	Description
Row ID	Integer	Unique sequential row identifier
Order ID	String	Unique order identifier (e.g., CA-2017-152156)
Order Date	Date	Date the order was placed (DD/MM/YYYY)
Ship Date	Date	Date the order was shipped
Ship Mode	Categorical	Shipping speed: Standard Class, Second Class, First Class, Same Day
Customer ID	String	Unique customer identifier (e.g., CG-12520)

Column	Type	Description
Customer Name	String	Full name of the customer
Segment	Categorical	Customer segment: Consumer, Corporate, Home Office
Country	String	Country of sale (always United States)
City	String	City where order was shipped
State	String	US state of delivery
Postal Code	String	Zip code of delivery address
Region	Categorical	US region: East, West, Central, South
Product ID	String	Unique product SKU identifier
Category	Categorical	Main product category
Sub-Category	Categorical	Product sub-category (17 types)
Product Name	String	Full name of the product
Sales	Float	Revenue generated by this line item (USD)

2. Product Catalog — Categories & Sub-Categories

RetailGPT covers 1,849 unique products organized into 3 main categories and 17 sub-categories. Products range from basic office stationery to high-value technology equipment and premium furniture.

Category: Furniture

The Furniture category encompasses all office and commercial furniture sold by the Superstore. It is the second-highest revenue category with $728,658 in total sales across 2,078 transactions. Average sale value is $350.65 — reflecting the higher price points of furniture items.

Sub-Categories:

Chairs — 607 orders | $322,822 total | $531.83 avg | 88 unique products
Office and task chairs including executive leather chairs, ergonomic task chairs, folding chairs, and conference room chairs. High-value sub-category.

Tables — 314 orders | $202,810 total | $645.89 avg | 56 unique products
Conference tables, adjustable-height desks, rectangular tables, and meeting room tables. Highest average sale value among Furniture sub-categories.

Bookcases — 226 orders | $113,813 total | $503.60 avg | 50 unique products
Wooden and laminate bookcases for office use; includes wall units and freestanding shelves.

Furnishings — 931 orders | $89,212 total | $95.82 avg | 186 unique products
Accessories and small furniture items: lamps, picture frames, clocks, rugs, curtains, coat racks. Most frequently ordered Furniture sub-category.

Category: Office Supplies

Office Supplies is the highest-volume category with 5,909 transactions — accounting for 60% of all orders. Total revenue is $705,422 with an average sale of $119.38. This category contains everyday consumables and office essentials that drive repeat purchasing.

Sub-Categories:

Binders — 1,492 orders | $200,028 total | $134.07 avg | 211 unique products
Ring binders, presentation binders, report covers. Includes brands like Avery, GBC, Cardinal, and Storex. Very high order frequency.

Paper — 1,338 orders | $76,828 total | $57.42 avg | 277 unique products
Largest product catalog with 277 SKUs. Includes copy paper, specialty paper, carbonless paper, photo paper. Low unit price but very high volume.

Storage — 832 orders | $219,343 total | $263.63 avg | 131 unique products
Filing cabinets, desk organizers, stackable drawers, hanging file folders, archival boxes, and personal safes.

Art — 785 orders | $26,705 total | $34.02 avg | 157 unique products
Pens, markers, pencils, colored pencils, correction fluid, dry-erase markers, highlighters, and art supplies.

Appliances — 459 orders | $104,618 total | $227.93 avg | 97 unique products
Electric kettles, coffee makers, fans, air purifiers, paper shredders, binding machines, and laminators.

Labels — 357 orders | $12,347 total | $34.59 avg | 70 unique products
Adhesive address labels, file folder labels, name badge labels, and label makers.

Supplies — 184 orders | $46,420 total | $252.28 avg | 36 unique products
Toner cartridges, ink cartridges, printer ribbons, and specialty office materials.

Envelopes — 248 orders | $16,128 total | $65.03 avg | 44 unique products
Business envelopes, padded mailers, bubble mailers, and security envelopes.

Fasteners — 214 orders | $3,001 total | $14.03 avg | 34 unique products
Rubber bands, binder clips, paper clips, push pins, and staple removers. Lowest average sale value overall.

Category: Technology

Technology is the highest-revenue category with $827,455 in total sales from 1,813 transactions. Average sale value is $456.40 — the highest of all three categories. Technology products include computing, communication, and electronic devices.

Sub-Categories:

Phones — 876 orders | $327,782 total | $374.18 avg | 189 unique products
Smartphones, desk phones, cordless phones, and phone accessories. Most ordered Technology sub-category.

Accessories — 756 orders | $164,186 total | $217.18 avg | 147 unique products
Computer peripherals: mice, keyboards, headsets, webcams, USB hubs, monitor stands, and cables.

Machines — 115 orders | $189,238 total | $1,645.55 avg | 63 unique products
High-value items: fax machines, label printers, binding equipment. Second highest average sale value per unit.

Copiers — 66 orders | $146,248 total | $2,215.88 avg | 13 unique products
Photocopiers and all-in-one printers. Fewest orders but HIGHEST average sale value at $2,215.88. Single copier order = significant revenue.

3. Sales Performance Analytics

Overall Sales Summary

Metric	Value
Total Revenue (all transactions)	$2,261,536.78
Total Number of Transactions	9,800
Unique Orders	4,922
Average Revenue per Transaction	$230.77
Median Revenue per Transaction	$54.49
Minimum Transaction Value	$0.4440
Maximum Transaction Value	$22,638.48
25th Percentile (Q1)	$17.25
75th Percentile (Q3)	$210.61
Standard Deviation of Sales	$626.65
Average Order Value (by Order ID)	$459.48

Sales by Category

Category	Total Sales ($)	Avg Sale ($)	Transactions	% of Revenue
Furniture	$728,658.58	$350.65	2,078	32.2%
Office Supplies	$705,422.33	$119.38	5,909	31.2%
Technology	$827,455.87	$456.40	1,813	36.6%

Technology leads in revenue ($827,455) despite having the fewest transactions (1,813), reflecting high-value unit pricing. Office Supplies dominates by transaction count (5,909 = 60% of all orders) but has a low average sale of $119. Furniture sits in the middle — high average value ($350) but moderate volume.

Sales by Sub-Category (All 17)

Sub-Category	Category	Total Sales ($)	Avg Sale ($)	Orders	Unique Products
Phones	Technology	$327,782.45	$374.18	876	189
Chairs	Furniture	$322,822.73	$531.83	607	88
Storage	Office Supplies	$219,343.39	$263.63	832	131

Sub-Category	Category	Total Sales ($)	Avg Sale ($)	Orders	Unique Products
Tables	Furniture	$202,810.63	$645.89	314	56
Binders	Office Supplies	$200,028.79	$134.07	1,492	211
Machines	Technology	$189,238.63	$1,645.55	115	63
Accessories	Technology	$164,186.70	$217.18	756	147
Copiers	Technology	$146,248.09	$2,215.88	66	13
Bookcases	Furniture	$113,813.20	$503.60	226	50
Appliances	Office Supplies	$104,618.40	$227.93	459	97
Furnishings	Furniture	$89,212.02	$95.82	931	186
Paper	Office Supplies	$76,828.30	$57.42	1,338	277
Supplies	Office Supplies	$46,420.31	$252.28	184	36
Art	Office Supplies	$26,705.41	$34.02	785	157
Envelopes	Office Supplies	$16,128.05	$65.03	248	44
Labels	Office Supplies	$12,347.73	$34.59	357	70
Fasteners	Office Supplies	$3,001.96	$14.03	214	34

Key Insights:

- Phones ($327,782) is the top revenue sub-category — highest total sales overall.
- Chairs ($322,822) is a close second and the top Furniture sub-category.
- Copiers have the highest average sale value at $2,215.88 per transaction.
- Machines average $1,645.55 per order — premium items with low order frequency.
- Fasteners are the cheapest sub-category at $14.03 average — consumables.
- Paper has the most SKUs (277 products) but very low average sale ($57.42).
- Tables have the fewest orders in Furniture (314) but highest Furniture avg ($645.89).
- Binders and Storage are the highest-volume Office Supplies sub-categories.

4. Regional & Geographic Intelligence

Sales by US Region

Region	Total Sales ($)	Avg Sale ($)	Orders	% of Revenue
Central	$492,646.91	$216.36	2,277	21.8%
East	$669,518.73	$240.40	2,785	29.6%
South	$389,151.46	$243.52	1,598	17.2%
West	$710,219.68	$226.18	3,140	31.4%

The West region leads in both total revenue ($710,219 = 31.6%) and transaction count (3,140 orders). The East is second at $669,518. Central and South trail significantly. Despite having the fewest transactions (1,598), the South achieves the highest average sale value ($243.52) among all regions.

Top 15 States by Revenue

Rank	State	Total Sales ($)	Region
1	California	$446,306.46	West
2	New York	$306,361.15	East
3	Texas	$168,572.53	Central
4	Washington	$135,206.85	West
5	Pennsylvania	$116,276.65	East
6	Florida	$88,436.53	South
7	Illinois	$79,236.52	Central
8	Michigan	$76,136.07	Central
9	Ohio	$75,130.35	East
10	Virginia	$70,636.72	South
11	North Carolina	$55,165.96	South
12	Indiana	$48,718.40	Central
13	Georgia	$48,219.11	South
14	Kentucky	$36,458.39	South
15	Arizona	$35,272.66	West

California dominates with $446,306 in revenue — nearly 1.5x more than second-place New York ($306,361). The top 3 states (California, New York, Texas) account for approximately 41% of total revenue. The top 5 states together generate over $1.17M (52% of total revenue).

Top 15 Cities by Revenue

Rank	City	State	Total Sales ($)
1	New York City	New York	$252,462.55
2	Los Angeles	California	$173,420.18
3	Seattle	Washington	$116,106.32
4	San Francisco	California	$109,041.12
5	Philadelphia	Pennsylvania	$108,841.75
6	Houston	Texas	$63,956.14
7	Chicago	Illinois	$47,820.13
8	San Diego	California	$47,521.03
9	Jacksonville	Florida	$44,713.18
10	Detroit	Michigan	$42,446.94
11	Springfield	Virginia	$41,827.81
12	Columbus	Ohio	$38,662.56
13	Newark	Ohio	$28,448.05
14	Columbia	South Carolina	$25,283.32
15	Jackson	Michigan	$24,963.86

New York City leads all cities at $252,462. Los Angeles ($173,420) and Seattle ($116,106) are strong second and third. The top 5 cities — New York City, Los Angeles, Seattle, San Francisco, and Philadelphia — together account for approximately $759,871 (34%) of total revenue.

Geographic Coverage Summary

- States covered: 49 US states
- Cities served: 529 unique cities
- All orders are domestic United States shipments only.
- West Region states include California, Washington, Oregon, Nevada, Arizona, Utah, Colorado, and others.
- East Region includes New York, Pennsylvania, Virginia, Florida, North Carolina, and New England states.
- Central Region covers Texas, Illinois, Ohio, Michigan, Wisconsin, Minnesota, and Midwest states.
- South Region includes Georgia, Tennessee, Alabama, Mississippi, Louisiana, Arkansas, and others.

5. Customer Segments & Profiles

The Superstore serves three distinct customer segments, each with different purchasing behaviors, average order values, and product preferences.

Segment Performance Overview

Segment	Total Sales ($)	Avg Sale ($)	Transactions	% Revenue	% Orders
Consumer	$1,148,060.53	$225.07	5,101	50.8%	52.1%
Corporate	$688,494.07	$233.15	2,953	30.4%	30.1%
Home Office	$424,982.18	$243.40	1,746	18.8%	17.8%

Segment: Consumer

The Consumer segment represents individual buyers — typically employees or professionals purchasing for personal or home use. With 5,101 transactions (52% of all orders) and $1.148M in revenue (51% of total), Consumers are the dominant customer type. They tend to purchase across all categories with a slight preference for Office Supplies due to frequency of small purchases. Average sale of $225 reflects a mix of small consumables and occasional high-value electronics.

Category	Revenue from This Segment
Technology	$401,011.66
Furniture	$387,696.26
Office Supplies	$359,352.61

Segment: Corporate

The Corporate segment represents businesses purchasing in bulk or for organizational needs. 2,953 transactions generate $688,494 in revenue (30.6% of total). Average sale of $233 is slightly higher than Consumer. Corporate clients often buy Technology and Furniture in larger quantities and may have procurement contracts. They are high-value retention targets.

Category	Revenue from This Segment
Technology	$244,041.84
Office Supplies	$224,130.54
Furniture	$220,321.70

Segment: Home Office

The Home Office segment serves remote workers and small business owners. With 1,746 transactions and $424,982 in revenue (18.9%), this is the smallest but highest average-value segment at $243 per transaction. Home Office buyers tend to invest in quality equipment, ergonomic furniture, and productivity tools. Growing segment aligned with remote work trends.

Category	Revenue from This Segment
Technology	$182,402.37
Office Supplies	$121,939.19
Furniture	$120,640.62

6. Shipping & Logistics

The Superstore offers four shipping modes with different speed-cost trade-offs. Understanding shipping patterns is essential for fulfillment planning, customer expectations, and SLA management.

Shipping Mode Summary

Ship Mode	Orders	Total Revenue	Avg Sale	Avg Days	Min Days	Max Days
First Class	1,501	$345,572.26	$230.23	2.2	1	4
Same Day	538	$125,219.04	$232.75	0.0	0	1
Second Class	1,902	$449,914.18	$236.55	3.2	1	5
Standard Class	5,859	$1,340,831.31	$228.85	5.0	3	7

Shipping Mode Details

Standard Class (59.8% of orders)

Delivery time: 3–7 business days (avg 5.0)

Cost level: Lowest cost

Default option for most non-urgent orders. Ground shipping. Most popular with 5,859 orders (59.8% of all shipments). Suitable for bulk office supplies and non-time-sensitive furniture.

Second Class (19.4% of orders)

Delivery time: 1–5 business days (avg 3.2)

Cost level: Low-medium cost

Faster than Standard Class. Used for moderate-priority shipments. 1,902 orders (19.4% of shipments). Good balance of speed and cost for corporate buyers with reasonable lead times.

First Class (15.3% of orders)

Delivery time: 1–4 business days (avg 2.2)

Cost level: Medium-high cost

Expedited shipping for urgent orders. 1,501 orders (15.3% of shipments). Frequently used for Technology items and time-sensitive supplies. Typically arrives within 2 business days.

Same Day (5.5% of orders)

Delivery time: 0–1 business days (avg 0.04)

Cost level: Highest cost

Premium same-day delivery. Only 538 orders (5.5% of shipments). Used for emergency office supply replenishment. Extremely fast but costs significantly more. Available in select cities only.

Shipping Insights

- Standard Class accounts for 59.8% of all orders — most customers prefer economy shipping.
- Same Day delivery is rare (5.5%) suggesting most orders are planned rather than emergency.
- Average shipping time across all modes is approximately 4.2 days.
- First Class and Same Day together represent only 20.8% of orders — premium shipping is a niche.

- All four shipping modes have similar average sale values ($228–$236) — price does not strongly correlate with shipping choice.
- Shipping time does not include processing/warehouse preparation time — only transit days.

7. Customer Intelligence — Top Buyers & Frequency

The dataset contains 793 unique customers. Customer analysis reveals high-value accounts and repeat buyers that represent the backbone of Superstore revenue.

Top 20 Customers by Total Revenue

Rank	Customer Name	Total Revenue	Avg per Transaction	Orders
1	Sean Miller	$25,043.05	$1,669.54	5
2	Tamara Chand	$19,052.22	$1,587.68	5
3	Raymond Buch	$15,117.34	$839.85	6
4	Tom Ashbrook	$14,595.62	$1,459.56	4
5	Adrian Barton	$14,473.57	$723.68	10
6	Ken Lonsdale	$14,175.23	$488.80	12
7	Sanjit Chand	$14,142.33	$642.83	9
8	Hunter Lopez	$12,873.30	$1,170.30	6
9	Sanjit Engle	$12,209.44	$642.60	11
10	Christopher Conant	$12,129.07	$1,102.64	5
11	Todd Sumrall	$11,891.75	$792.78	6
12	Greg Tran	$11,820.12	$407.59	11
13	Becky Martin	$11,789.63	$736.85	4
14	Seth Vernon	$11,470.95	$358.47	10
15	Caroline Jumper	$11,164.97	$558.25	8
16	Clay Ludtke	$10,880.55	$388.59	12
17	Maria Etezadi	$10,663.73	$484.71	10
18	Karen Ferguson	$10,604.27	$589.13	7
19	Bill Shonely	$10,501.65	$1,166.85	5
20	Edward Hooks	$9,940.38	$368.16	10

Sean Miller is the top customer at $25,043 — driven largely by a single high-value Technology order (CA-2015-145317, $22,638). Tamara Chand and Raymond Buch follow with $19,052 and $15,117 respectively. The top 20 customers together account for approximately $239,000 in revenue.

Most Frequently Ordering Customers

Customer Name	Transactions	Segment
William Brown	35	Consumer
Paul Prost	34	Home Office
Matt Abelman	34	Home Office
John Lee	33	Consumer
Seth Vernon	32	Consumer
Chloris Kastensmidt	32	Consumer
Jonathan Doherty	32	Corporate
Arthur Prichep	31	Consumer
Emily Phan	31	Consumer
Zuschuss Carroll	31	Consumer
Lena Cacioppo	30	Consumer
Brian Moss	29	Corporate
Dean percer	29	Home Office
Ken Lonsdale	29	Consumer
Greg Tran	29	Consumer

William Brown, Matt Abelman, and Paul Prost each have 34–35 transactions — they are the most active repeat buyers. High transaction frequency indicates loyal customers with ongoing supply needs.

8. Product Intelligence — Best Sellers & High Value

Most Frequently Ordered Products (Top 20)

Rank	Product Name	Orders	Sub-Category
1	Staple envelope	47	Envelopes
2	Staples	46	Fasteners
3	Easy-staple paper	44	Paper
4	Avery Non-Stick Binders	20	Binders
5	Staple remover	18	Supplies
6	Staples in misc. colors	18	Art
7	Storex Dura Pro Binders	17	Binders
8	KI Adjustable-Height Table	17	Tables
9	Staple-based wall hangings	16	Furnishings
10	Logitech 910-002974 M325 Wireless Mouse for Web Scrolli	15	Accessories
11	Situations Contoured Folding Chairs, 4/Set	15	Chairs
12	GBC Premium Transparent Covers with Diagonal Lined Patt	14	Binders
13	Eldon Wave Desk Accessories	14	Furnishings
14	Global High-Back Leather Tilter, Burgundy	14	Chairs
15	Global Wood Trimmed Manager's Task Chair, Khaki	14	Chairs
16	SAFCO Arco Folding Chair	13	Chairs
17	Staple holder	13	Appliances
18	Hot File 7-Pocket, Floor Stand	13	Storage
19	Chromcraft Round Conference Tables	13	Tables
20	GBC Instant Report Kit	13	Binders

Staple envelope (47 orders), Staples (46), and Easy-staple paper (44) dominate order frequency. These are low-cost consumables that customers repurchase regularly. The most ordered products are predominantly Office Supplies — confirming the high-volume, low-value nature of that category.

Highest Single-Transaction Revenue Products

Rank	Product	Customer	Revenue
1	Cisco TelePresence System EX90 Videoconferenc	Sean Miller	$22,638.48
2	Canon imageCLASS 2200 Advanced Copier	Tamara Chand	$17,499.95
3	Canon imageCLASS 2200 Advanced Copier	Raymond Buch	$13,999.96
4	Canon imageCLASS 2200 Advanced Copier	Tom Ashbrook	$11,199.97
5	Canon imageCLASS 2200 Advanced Copier	Hunter Lopez	$10,499.97
6	GBC Ibimaster 500 Manual ProClick Binding Sys	Adrian Barton	$9,892.74
7	Ibico EPK-21 Electric Binding System	Sanjit Chand	$9,449.95
8	3D Systems Cube Printer, 2nd Generation, Mage	Bill Shonely	$9,099.93
9	HP Designjet T520 Inkjet Large Format Printer	Sanjit Engle	$8,749.95
10	Canon imageCLASS 2200 Advanced Copier	Christopher Conant	$8,399.98
11	High Speed Automatic Electric Letter Opener	Ken Lonsdale	$8,187.65
12	Lexmark MX611dhe Monochrome Laser Printer	Becky Martin	$8,159.95
13	Cubify CubeX 3D Printer Triple Head Print	Grant Thornton	$7,999.98
14	HP Designjet T520 Inkjet Large Format Printer	Tom Boeckenhauer	$6,999.96
15	Fellowes PB500 Electric Punch Plastic Comb Bi	Christopher Martinez	$6,354.95

Revenue by Product (Top 15 Products)

Rank	Product Name	Sub-Category	Total Revenue
1	Canon imageCLASS 2200 Advanced Copier	Copiers	$61,599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding	Binders	$27,453.38
3	Cisco TelePresence System EX90 Videoconferencing U	Machines	$22,638.48
4	HON 5400 Series Task Chairs for Big and Tall	Chairs	$21,870.58
5	GBC DocuBind TL300 Electric Binding System	Binders	$19,823.48
6	GBC Ibimaster 500 Manual ProClick Binding System	Binders	$19,024.50
7	Hewlett Packard LaserJet 3310 Copier	Copiers	$18,839.69
8	HP Designjet T520 Inkjet Large Format Printer - 24	Machines	$18,374.90
9	GBC DocuBind P400 Electric Binding System	Binders	$17,965.07
10	High Speed Automatic Electric Letter Opener	Supplies	$17,030.31
11	Lexmark MX611dhe Monochrome Laser Printer	Machines	$16,829.90
12	Martin Yale Chadless Opener Electric Letter Opener	Supplies	$16,656.20

Rank	Product Name	Sub-Category	Total Revenue
13	Ibico EPK-21 Electric Binding System	Binders	$15,875.92
14	Riverside Palais Royal Lawyers Bookcase, Royale Ch	Bookcases	$15,610.97
15	3D Systems Cube Printer, 2nd Generation, Magenta	Machines	$14,299.89

9. Data Versions & Synthetic Datasets

The RetailGPT system uses three data files: the original training dataset (train.csv) and two synthetically generated balanced versions used for model training and evaluation.

Original Training Data (train.csv)

The primary Superstore dataset. Contains real transactional data with natural class imbalance — Office Supplies dominates at 60% of orders. Sales distribution is right-skewed with median $54.49 and outliers reaching $22,638. Used as the ground truth for analytics and RAG retrieval.

Property	Value
File Name	train.csv
Total Rows	9,800
Mean Sale	$230.77
Median Sale	$54.49
Columns	18 (same schema as train.csv)

Synthetic Balanced Version 1 (synthetic_balanced_v1.csv)

Synthetically generated dataset with balanced sub-category representation. Significantly higher mean sales ($2,044) suggesting upsampling of high-value transactions. Used for training ML models that require balanced class distribution. Paper (1,441), Binders (1,111), and Phones (1,015) are the most represented sub-categories.

Property	Value
File Name	synthetic_balanced_v1.csv
Total Rows	9,800
Mean Sale	$2,044.41
Median Sale	$207.02
Columns	18 (same schema as train.csv)

Synthetic Balanced Version 2 (synthetic_balanced_v2.csv)

Second synthetic version with even higher average sales ($2,770) and median ($485.68). Represents a more aggressive upsampling of high-value orders. Higher standard deviation suggests greater variability. Used as a complementary training dataset for robustness testing.

Property	Value
File Name	synthetic_balanced_v2.csv
Total Rows	9,800
Mean Sale	$2,770.81
Median Sale	$485.68
Columns	18 (same schema as train.csv)

Dataset Schema Consistency

All three datasets share an identical 18-column schema. This consistency ensures that any ETL pipeline, ML model, or RAG ingestion process built for train.csv works without modification on the synthetic datasets. The only differences are in value distributions and potential synthetic Customer IDs and Order IDs in the balanced versions.

10. System Architecture — RetailGPT Azure Deployment

RetailGPT is designed for enterprise deployment on Microsoft Azure. The architecture separates concerns across Frontend, Backend API, Identity/Security, Data Engineering, and Multi-Agent AI layers.

Component Mapping: Local → Azure

Local Component	Azure Native Service	Rationale
FastAPI Backend	Azure App Service (Web App for Containers)	Serverless scaling, VNet integration, Docker support
Streamlit Frontend	Azure App Service	Separate tier decouples UI scaling from API load
SQLite Database	Azure SQL Database	Relational integrity, automated backups, enterprise compliance
Pandas Pipeline	Azure Databricks + Data Factory	Databricks handles PySpark workloads; ADF orchestrates triggers
File Storage (local)	Azure Data Lake Storage Gen2	Hierarchical namespace: Raw, Staged, Curated zones
Power BI CSV Export	Microsoft Fabric (DirectLake)	Power BI connects directly to Parquet files in OneLake
LangChain + Groq LLM	Azure OpenAI Service	Enterprise SLAs, data privacy, RBAC controls
ChromaDB (Local RAG)	Azure AI Search	Built-in semantic ranking, hybrid search, Azure OpenAI integration
Local bcrypt secrets	Azure Key Vault	Centralized secret management with role-based access
Local Docker build	Azure Container Registry (ACR)	Secure private image registry integrated with App Service

Data Engineering Pipeline

Step 1 — Ingest: Raw CSV files (train.csv, synthetic versions) are uploaded to Azure Data Lake Gen2 in the raw/ container.

Step 2 — Orchestrate: Azure Data Factory triggers a pipeline on schedule or event (new file arrival).

Step 3 — Transform: Azure Databricks runs PySpark jobs to clean, validate, and enrich data — handling null values, type casting, date parsing, and feature engineering.

Step 4 — Store: Curated Parquet files are written to the curated/ container in ADLS Gen2 with optimized partitioning (by region/year/category).

Step 5 — Serve: Microsoft Fabric connects via DirectLake to Parquet files — enabling real-time Power BI dashboards without data movement.

Multi-Agent AI Architecture

The AI layer uses LangChain as the orchestration framework coordinating multiple specialized agents. The Retrieval-Augmented Generation (RAG) system uses Azure AI Search as the vector store, indexed with product knowledge, policy documents, and historical order data. Azure OpenAI GPT-4o is the inference engine powering all natural language responses.

- **Query Router Agent:** Classifies incoming user questions and routes them to the appropriate specialized agent or data source.
- **Analytics Agent:** Handles quantitative questions — sales figures, aggregations, comparisons, and trend analysis.
- **Product Agent:** Answers product catalog questions — descriptions, categories, pricing, and availability.
- **Customer Agent:** Handles customer profile lookups, order history, and segment-level insights.
- **Logistics Agent:** Answers shipping-related questions — delivery times, ship modes, and regional policies.
- **RAG Knowledge Agent:** Retrieves from the vector store (this document) for policy, architecture, and general knowledge questions.

CI/CD Pipeline

- GitHub Actions builds the Docker image on every push to main branch.
- Image is pushed to Azure Container Registry (ACR) with versioned tags.
- A webhook is triggered on Azure App Service to pull the latest image automatically.
- Environment variables and API keys are stored in Azure Key Vault and injected at runtime.
- Database migrations are run automatically as part of the deployment pipeline.

Migration Checklist

- ■ Provision Azure SQL Database and run migration script for users and uploads tables.
- ■ Upload train.csv to ADLS Gen2 raw/ container.
- ■ Migrate data_pipeline.py logic to a Databricks Notebook.
- ■ Provision Azure OpenAI and update .env or Key Vault with AZURE_OPENAI_API_KEY.
- ■ Push local Docker image to Azure Container Registry.
- ■ Connect Power BI Service directly to the Microsoft Fabric workspace.
- ■ Configure Azure AI Search index with product and knowledge base documents.
- ■ Set up Azure Key Vault references in App Service configuration.

11. Frequently Asked Questions (100+ Questions)

This section covers the most likely questions users will ask the RetailGPT chatbot, organized by topic. Answers are grounded in actual dataset statistics.

General Business Questions

Q: What is this retail store called?

A: The store is referred to as the Superstore — a fictional US-based retail company used for business intelligence and analytics training. It sells products in three main categories: Furniture, Office Supplies, and Technology.

Q: What does this store sell?

A: The Superstore sells products in three categories: (1) Furniture — chairs, tables, bookcases, furnishings; (2) Office Supplies — binders, paper, storage, art supplies, appliances, labels, envelopes, fasteners; (3) Technology — phones, accessories, machines, copiers.

Q: How many products does the store carry?

A: The store carries 1,849 unique products across 17 sub-categories.

Q: What is the total revenue of the store?

A: Total revenue from the dataset is $2,261,536.78 across 9,800 transactions.

Q: How many orders have been placed?

A: There are 4,922 unique orders in the dataset, from a total of 9,800 transaction line items.

Q: How many customers does the store have?

A: The store has 793 unique customers in the dataset.

Q: In which country does the store operate?

A: The Superstore operates exclusively in the United States, serving all four major US geographic regions.

Q: What time period does the data cover?

A: The dataset covers sales from 2015 through 2018, spanning approximately 4 years of transactions.

Q: What is the average order value?

A: The average revenue per transaction line item is $230.77. The average total order value (sum of all items per order) is approximately $459.48.

Q: What was the highest single sale ever?

A: The highest single transaction was $22,638.48 — a Technology product ordered by Sean Miller (Order ID: CA-2015-145317).

Product & Category Questions

Q: What are the main product categories?

A: The three main product categories are: (1) Furniture, (2) Office Supplies, and (3) Technology.

Q: Which category generates the most revenue?

A: Technology generates the most revenue at $827,455 (36.8% of total), despite having the fewest transactions (1,813). Office Supplies is second at $705,422, and Furniture third at $728,658.

Q: Which category has the most orders?

A: Office Supplies has by far the most orders — 5,909 transactions, representing 60.3% of all orders. It is the high-volume, lower-value category.

Q: What is the most expensive product category?

A: Technology has the highest average sale value at $456.40 per transaction, making it the most expensive category on average.

Q: What are all 17 sub-categories?

A: The 17 sub-categories are: Bookcases, Chairs, Furnishings, Tables (Furniture); Art, Binders, Envelopes, Fasteners, Labels, Paper, Storage, Supplies, Appliances (Office Supplies); Accessories, Copiers, Machines, Phones (Technology).

Q: Which sub-category generates the most revenue?

A: Phones generates the most sub-category revenue at $327,782. Chairs is close behind at $322,822.

Q: Which sub-category has the highest average sale?

A: Copiers have the highest average sale at $2,215.88 per transaction, followed by Machines at $1,645.55.

Q: Which sub-category is ordered most frequently?

A: Binders (1,492 orders) and Paper (1,338 orders) are the most frequently ordered sub-categories.

Q: What is the cheapest sub-category on average?

A: Fasteners are the cheapest, averaging only $14.03 per transaction.

Q: How many furniture products are available?

A: Furniture contains approximately 550 unique products across Chairs (88), Tables (56), Bookcases (50), and Furnishings (186 products).

Q: What types of chairs are sold?

A: The Chairs sub-category includes executive leather chairs, ergonomic task chairs, folding chairs, conference room chairs, and drafting chairs from brands like Global, Hon, and Situation.

Q: What technology products are sold?

A: Technology products include smartphones, desk phones, cordless phones, computer accessories (mice, keyboards, headsets, webcams), copiers, fax machines, label printers, and binding equipment.

Q: What is the most ordered product?

A: Staple envelope is the most frequently ordered product with 47 orders. Staples (46) and Easy-staple paper (44) are close second and third.

Q: How many sub-categories does Office Supplies have?

A: Office Supplies has 9 sub-categories: Art, Binders, Envelopes, Fasteners, Labels, Paper, Storage, Supplies, and Appliances.

Q: Which product has the most SKUs available?

A: Paper has the widest selection with 277 unique product variants, followed by Binders (211 products) and Phones (189 products).

Sales & Revenue Questions

Q: What is the total revenue?

A: Total revenue is $2,261,536.78 across all 9,800 transaction records.

Q: What is the median sale value?

A: The median (50th percentile) sale value is $54.49.

Q: What is the minimum sale value ever recorded?

A: The minimum sale value is $0.4440.

Q: What is the maximum sale value ever recorded?

A: The maximum sale value is $22,638.48.

Q: Which region has the highest sales?

A: The West region leads with $710,219.68 in total revenue (31.6% of total), followed by the East at $669,518.73.

Q: Which region has the most orders?

A: The West has the most orders with 3,140 transactions (32.0%), followed by the East with 2,785 orders.

Q: Which state generates the most revenue?

A: California is the top state at $446,306 in revenue — nearly 1.5x more than second-place New York ($306,361).

Q: What are the top 5 states by sales?

A: Top 5 states: (1) California $446,306, (2) New York $306,361, (3) Texas $168,572, (4) Washington $135,207, (5) Pennsylvania $116,277.

Q: Which city has the highest sales?

A: New York City leads all cities with $252,462 in total revenue, followed by Los Angeles at $173,420.

Q: What percentage of revenue comes from Technology?

A: Technology accounts for approximately 36.8% of total revenue ($827,455.87 of $2,261,536.78).

Q: What percentage of orders are Office Supplies?

A: Office Supplies accounts for 60.3% of all transaction orders (5,909 out of 9,800).

Q: How much does the average customer spend?

A: Across all customers, total revenue is distributed across 793 customers, averaging approximately $2,851.87 per customer.

Q: What is the revenue split by segment?

A: Consumer: $1,148,061 (51.1%), Corporate: $688,494 (30.6%), Home Office: $424,982 (18.9%).

Customer Questions

Q: Who is the top customer by revenue?

A: Sean Miller is the top customer with $25,043 in total purchases — primarily from a single large Technology order ($22,638).

Q: Who are the top 5 customers?

A: Top 5 customers by revenue: (1) Sean Miller $25,043, (2) Tamara Chand $19,052, (3) Raymond Buch $15,117, (4) Tom Ashbrook $14,596, (5) Adrian Barton $14,474.

Q: How many customer segments are there?

A: There are three customer segments: Consumer, Corporate, and Home Office.

Q: Which segment has the most customers?

A: The Consumer segment has the most transactions (5,101 = 52%) and represents the largest customer group.

Q: Which segment spends the most per transaction?

A: Home Office has the highest average transaction value at $243.40, slightly above Corporate ($233.15) and Consumer ($225.07).

Q: Who orders most frequently?

A: William Brown orders most frequently with 35 transactions, followed by Matt Abelman and Paul Prost (34 each).

Q: How many unique customers are there?

A: There are 793 unique customers in the dataset.

Q: What is a Corporate segment customer?

A: Corporate customers are businesses purchasing for organizational needs. They represent 30.6% of revenue and tend to make slightly higher-value purchases than Consumer segment.

Q: What is a Home Office customer?

A: Home Office customers are remote workers and small business owners. They make up 18.9% of revenue and have the highest average purchase value ($243) among segments.

Shipping & Logistics Questions

Q: What shipping options are available?

A: Four shipping modes are available: Standard Class (economy), Second Class (standard), First Class (expedited), and Same Day (premium instant delivery).

Q: How long does Standard Class shipping take?

A: Standard Class takes 3–7 business days, with an average of 5.0 days.

Q: How long does First Class shipping take?

A: First Class takes 1–4 business days, with an average of 2.2 days.

Q: How long does Same Day shipping take?

A: Same Day delivery is completed the same day or within 1 business day (average 0.04 days from order to ship).

Q: How long does Second Class shipping take?

A: Second Class takes 1–5 business days, with an average of 3.2 days.

Q: What is the most popular shipping method?

A: Standard Class is most popular with 5,859 orders (59.8% of all shipments).

Q: How many orders use Same Day shipping?

A: Only 538 orders (5.5% of all orders) use Same Day shipping — it is the least common option.

Q: Is shipping available across all US states?

A: Yes, the Superstore ships to 49 US states and 529 cities nationwide.

Q: Does shipping mode affect what I can buy?

A: No — all shipping modes are available for all products. The choice affects delivery speed and cost, not product eligibility.

Q: What is the fastest shipping option?

A: Same Day is the fastest option, completing delivery the same day as order placement in most cases.

Technical / System Questions

Q: What is RetailGPT?

A: RetailGPT is an AI-powered retail intelligence system built with LangChain, FastAPI, Streamlit, and Azure OpenAI. It answers natural language questions about Superstore sales data using a multi-agent RAG architecture.

Q: What AI model powers RetailGPT?

A: RetailGPT uses Azure OpenAI GPT-4o as the primary inference model, with LangChain as the orchestration framework and Azure AI Search as the vector database for RAG retrieval.

Q: What is a RAG system?

A: RAG (Retrieval-Augmented Generation) is an AI architecture that retrieves relevant documents from a knowledge base and injects them into the LLM prompt to produce grounded, accurate answers.

Q: Where is the data stored?

A: In production Azure deployment, data is stored in Azure Data Lake Gen2 as Parquet files. Customer/user data is in Azure SQL Database. The vector index is in Azure AI Search.

Q: How is user authentication handled?

A: Locally, SQLite stores users with bcrypt password hashing. In Azure production, users are stored in Azure SQL Database and secrets managed via Azure Key Vault.

Q: What database does RetailGPT use?

A: The application uses SQLite locally for development and Azure SQL Database in production. Data is served from Azure Data Lake (Parquet) for analytics.

Q: How is the data pipeline structured?

A: The pipeline flows: Raw CSV → Azure Data Lake Gen2 raw/ → Azure Data Factory orchestration → Azure Databricks transformation → Curated Parquet → Microsoft Fabric / Power BI.

Q: What is the frontend built with?

A: The frontend is built with Streamlit, deployed as an Azure App Service container.

Q: What is the backend built with?

A: The backend uses FastAPI with multiple routers, containerized with Docker and deployed on Azure App Service.

Q: How does the multi-agent system work?

A: A query router agent classifies incoming questions and routes them to specialized agents: Analytics Agent, Product Agent, Customer Agent, Logistics Agent, and RAG Knowledge Agent.

Q: What datasets are used for training?

A: Three CSV files: train.csv (original 9,800 rows), synthetic_balanced_v1.csv (9,800 rows, balanced), and synthetic_balanced_v2.csv (9,800 rows, higher-value synthetic data).

Q: What is synthetic_balanced_v1.csv?

A: synthetic_balanced_v1.csv is a synthetically generated dataset with 9,800 rows designed to balance sub-category representation. It has a higher average sale ($2,044) than the original train.csv ($230).

Q: What is the difference between train.csv and synthetic versions?

A: train.csv is the original real-world dataset with natural class imbalance (60% Office Supplies). Synthetic versions artificially balance sub-categories for better ML model training.

Q: How is the Azure deployment done?

A: GitHub Actions builds Docker images, pushes to Azure Container Registry, and triggers automatic deployment via webhook to Azure App Service.

Q: What is DirectLake in the architecture?

A: DirectLake is a Microsoft Fabric feature that allows Power BI to connect directly to Parquet files in OneLake (ADLS Gen2) without importing data — enabling real-time analytics at scale.

Analytical & Business Intelligence Questions

Q: What is the most profitable sub-category?

A: Without profit margin data, we use revenue as a proxy. Phones ($327,782) generates the most sub-category revenue. However, Copiers have the highest value per sale ($2,215) suggesting high margins.

Q: What products are frequently bought together?

A: While basket analysis requires transaction-level join analysis, Staple envelopes, Staples, and Easy-staple paper are frequently co-ordered — typical office supply replenishment bundles.

Q: What are the highest-risk customer accounts to lose?

A: Top customers Sean Miller ($25,043), Tamara Chand ($19,052), and Raymond Buch ($15,117) represent highest revenue-at-risk. High-frequency buyers like William Brown (35 orders) are also critical for recurring revenue.

Q: Which region is underperforming?

A: The South region has the fewest orders (1,598) and lowest total revenue ($389,151 = 17.3%) despite having the highest average sale ($243.52). This suggests fewer customers but high-value purchases when they occur.

Q: What is the revenue concentration risk?

A: California alone accounts for 19.8% of total revenue. The top 2 states (California and New York) contribute 33.5% of revenue. High geographic concentration is a business risk.

Q: How does Home Office segment differ from Consumer?

A: Home Office customers have a higher average purchase value ($243 vs $225) but represent only 18.9% of revenue vs 51.1% for Consumer. Home Office buyers make fewer but higher-value purchases.

Q: Are there seasonal sales patterns?

A: Without aggregating by month/quarter, the dataset covers 2015–2018. Typical retail seasonal patterns (Q4 holiday peaks) would need time-series analysis to confirm in this dataset.

Q: What is the customer lifetime value metric?

A: Average customer lifetime value in this dataset is approximately $2,851.87 (total revenue ÷ unique customers). Top customers like Sean Miller far exceed this at $25,043.

Q: Which products have the highest unit value?

A: Copiers average $2,215.88 per order, Machines $1,645.55, Tables $645.89, and Chairs $531.83 are the highest unit value sub-categories.

Q: What percentage of revenue comes from the top 10 customers?

A: Top 10 customers together contribute approximately $152,225 out of $2,261,536.78 total — roughly 6.8% of all revenue from just 10 customers.

Operational Questions

Q: How do I filter orders by region?

A: The Region column contains four values: East, West, Central, South. Filter by this column to analyze regional performance.

Q: How do I identify high-value orders?

A: Sort or filter the Sales column for values above $1,000. The top single transactions are in the Technology category (Copiers, Machines, Phones).

Q: Can I search by customer name?

A: Yes — the Customer Name column contains full names (e.g., "Sean Miller"). Customer IDs follow the format XX-##### (e.g., CG-12520).

Q: How are orders identified?

A: Orders use the format XX-YYYY-######. CA prefix means California (domestic), US prefix may indicate a different origin state. Example: CA-2017-152156.

Q: How are products identified?

A: Products have a Product ID in format CAT-SUBCAT-######## (e.g., FUR-BO-10001798 = Furniture-Bookcases). Product Names are full descriptive strings.

Q: What date format is used?

A: Dates are in DD/MM/YYYY format (day first). Example: 08/11/2017 = November 8, 2017.

Q: How many states does the store ship to?

A: The store ships to 49 states across the continental United States.

Q: What is the postal code format?

A: Postal codes are standard US ZIP codes (5-digit numeric). Some synthetic dataset versions may include decimal points (e.g., 49505.0) which should be cleaned.

Q: Can I filter by product category?

A: Yes — the Category column contains exactly three values: Furniture, Office Supplies, Technology. The Sub-Category column provides 17 more granular groupings.

Q: What is the Row ID field?

A: Row ID is a simple sequential integer (1, 2, 3...) that uniquely identifies each transaction row. It is not an order or product identifier.

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12. Glossary of Terms

Term	Definition
RAG	Retrieval-Augmented Generation — AI architecture that combines vector search with LLM generation for grounded answe
LangChain	Python framework for building LLM-powered applications with chains and agents.
Azure OpenAI	Microsoft Azure-hosted version of OpenAI GPT models with enterprise SLAs and data privacy.
Azure AI Search	Microsoft Azure vector database and semantic search service, used as the RAG vector store.
ChromaDB	Open-source local vector database used in development mode before Azure AI Search deployment.
FastAPI	High-performance Python web framework used for the RetailGPT backend API.
Streamlit	Python framework for building interactive web applications, used as the RetailGPT frontend.
ADLS Gen2	Azure Data Lake Storage Generation 2 — hierarchical cloud storage for big data workloads.
Azure Databricks	Managed Apache Spark service on Azure used for large-scale data transformation.
Azure Data Factory	Cloud ETL/ELT orchestration service that triggers and manages data pipelines.
Microsoft Fabric	Unified analytics platform from Microsoft combining data engineering, warehousing, and Power BI.
DirectLake	Fabric/Power BI mode that reads directly from Delta/Parquet files in OneLake without import.
Transaction	A single line item in an order — one product purchased at a specific quantity and price.
Order	A group of transactions (line items) placed together with a single Order ID.
SKU	Stock Keeping Unit — a unique identifier for a product variant (mapped to Product ID here).
Segment	Customer classification: Consumer, Corporate, or Home Office based on buyer type.
Sub-Category	A specific product type within a category (e.g., Chairs within Furniture).
Ship Mode	The delivery speed option selected: Standard Class, Second Class, First Class, Same Day.
Region	US geographic division: East, West, Central, or South.
bcrypt	Password hashing algorithm used to securely store user passwords.
Azure Key Vault	Azure service for securely storing and accessing secrets, keys, and certificates.
ACR	Azure Container Registry — private Docker image registry for containerized deployments.
CI/CD	Continuous Integration / Continuous Deployment — automated build, test, and deploy pipeline.
Vector Store	Database optimized for storing and querying high-dimensional embedding vectors for semantic search.
Embedding	Numerical vector representation of text that captures semantic meaning for similarity search.
GPT-4o	OpenAI's multimodal language model used as the inference engine for RetailGPT responses.
Parquet	Columnar storage file format optimized for analytics workloads; used in the curated data layer.
PySpark	Python API for Apache Spark, used in Databricks for large-scale parallel data processing.

Term	Definition
Webhook	HTTP callback mechanism used to trigger Azure App Service redeployment after ACR image push.